Association between manganese superoxide dismutase (MnSOD) Val-9Ala polymorphism and cancer risk - A meta-analysis.

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A growing body of evidence suggests that reactive oxygen species (ROS) play an important role in human cancers. Manganese superoxide dismutase (MnSOD) is the major antioxidant in the mitochondria, catalysing the dismutation of superoxide radicals to form hydrogen peroxide. Since the identification of a well-characterised functional polymorphism, Val-9Ala of MnSOD, a number of molecular epidemiological studies have evaluated the association between Val-9Ala and cancer risk. However, the results remain conflicting rather than conclusive. This meta-analysis on 15,320 cancer cases and 19,534 controls from 34 published case-control studies shows no significant overall main effect of MnSOD Val-9Ala on cancer risk. However, we found that the MnSOD 9Ala allele was associated with an increased prostate cancer risk (Val/Ala versus Val/Val: odds ratio (OR)=1.1; 95% confidence intervals (CI): 1.0-1.3; Ala/Ala versus Val/Val: OR=1.3; 95% CI: 1.0-1.6; Val/Ala+Ala/Ala versus Val/Val: OR=1.2; 95% CI, 1.0-1.3). In addition, we found that the MnSOD Ala-9Ala genotype contributed to an increased breast cancer risk in premenopausal women who had low consumption of antioxidants (Ala/Ala versus Val/Ala+Val/Val: OR=2.6, 95% CI: 1.0-6.4 with low vitamin C consumption; OR=2.1, 95%CI: 1.3-3.4 with low vitamin E consumption and OR=2.9, 95%CI: 1.5-5.7 with low carotenoid consumption). These results suggest that the MnSOD Val-9Ala polymorphism may contribute to cancer development through a disturbed antioxidant balance.